DGFNet: Depth-Guided Cross-Modality Fusion Network for RGB-D Salient Object Detection

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***Abstract*—RGB-D salient object detection (SOD) focuses on utilizing the complementary cues of RGB and depth modalities to detect and segment salient regions. However, many proposed methods train their models in a simple multi-modal manner, ignoring the differences between these two modalities in the contribution of salient detection. Furthermore, the quality of depth datasets varies significantly between individuals and is another important factor affecting model performance. To address the aforementioned issues, this article proposes a novel depth-guided fusion network framework (DGFNet) for the RGB-D SOD task. To avoid the influence of low-quality depth maps on RGB-D SOD, we design a depth map enhanced algorithm which jointly models salient detection and depth estimation to improve the quality of depth. Also, we propose a depth attention mechanism to encode valuable spatial information for SOD, which is then used in depth-guided fusion (DGF) module to guide the fusion of cross-modality features at each level. Extensive experiments on seven commonly tested datasets demonstrate that our DGFNet outperforms the 23 state-of-the-art RGB-D-based SOD methods.**

***Index Terms*—RGB-D salient object detection, depth quality, cross-modal feature fusion.**

I. Introduction

SALIENT object detection (SOD) aims to discover the most visually attractive object or region in a scene, and segments them from the background. It has been widely used in many related computer vision tasks, including image recognition [1], object detection [2], image segmentation [3], image quality assessment [4], image retrieval [5], visual tracking [6], image understanding [7], etc. However, in some applications captured RGB images often suffer from low contrast, color distortion, poor visibility, and other problems, which bring challenges to detecting the salient object for such complex scenes. As a complementary of RGB data, depth information can be easily acquired by low-cost depth sensors [8] and has become a practical option for high-quality salient object detection. The corresponding RGB-D SOD method has received increasing interest with promising progress.

Although traditional RGB-D salient object detection [9], [10], [11] extracted the more correlated and diversified handcrafted features, it could not generalize well on unseen data. With the rapid development of deep learning, many convolutional neural networks (CNNs) have been proposed and achieved good results on salient object detection due to their good feature extraction capabilities [12], [13], [14], [15], [16], [17]. Existing CNN-based work mainly focuses on extracting multimodality features and integrating multiple complementary information in different levels. Specifically, input-side fusion [18], [19] uses a depth map as an additional channel alongside the RGB inputs to infer the salient object. The output-side fusion [20], [21], [22] utilizes two CNNs to predict the results from the RGB image and depth map separately, and fuse the predictions to produce the final results. Different from the above methods, the feature-level fusion [23], [24], [25], [26], [27], [28], [29] learns fusion features from the RGB and depth branch streams, which can retain more semantic information and detect salient object more accurately.

Most of existing fusion methods indiscriminately treat the information from different sources with the same operation during the fusion process. In fact, the characteristics of cross-modal information are different, RGB images provide texture and color information, but the saliency detection is easily confused by similar texture and complex backgrounds [13]. As shown in Fig. 1(a), the depth map has rich spatial information and keeps more in accordance with salient objects. Efficient depth features may be more helpful for locating and segmenting object. Inspired by the above observations, the depth-guided fusion (DGF) module is proposed to fuse the RGB and Depth features. The DGF module focuses on leverage the spatial information of depth data by attention and weighting mechanisms to guide the representation of two-modality features on saliency regions. In this way, the spatial representation of each modality feature is augmented to highlight saliency cues. More important, it can suppress irrelevant features for SOD to reduce the gap between

Manuscript received 26 February 2023; revised 9 June 2023; accepted 19 July 2023. Date of publication 2 August 2023; date of current version 2 February 2024. This work was supported in part by the National Science and Technology Major Project under Grant 2020YFA0713504, in part by the National Natural Science Foundation of China under Grant 61972333, and in part by the Scientific Research Foundation of Education Department of Hunan Province under Grant 21A0109. The Associate Editor coordinating the review of this manuscript and approving it for publication was Mr. Wu Liu. *(Fen Xiao and Zhengdong Pu contributed equally to this work.) (Corresponding author: Fen Xiao; Xieping Gao.)*

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Digital Object Identifier 10.1109/TMM.2023.3301280

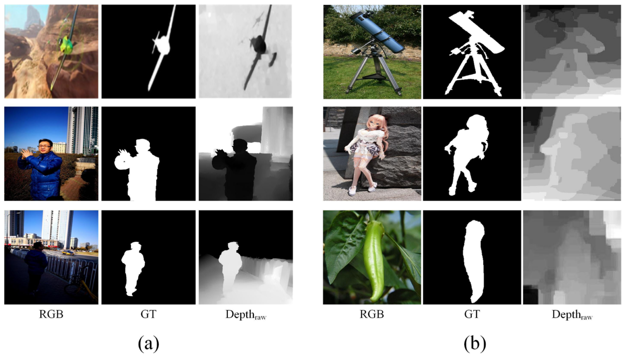

Fig. 1. Some examples of RGB images, Depth maps and GT.

modalities, thus effectively fusing RGB and Depth information.

On the other hand, as shown in Fig. 1(b), the depth map is easy to be noised by hardware or the visual scene, which causes incomplete or erroneous spatial structure. The introduction of such depth maps with low quality will affect the performance of SOD. To overcome this problem, researchers proposed many approaches to choose or upgrade low-quality depth information. Chen et al. [30] acquired additional depth information by the retrieving of a small set of similar images from external datasets, and some methods [31], [32], [33] discard or give very small weights to low-quality depth map features in the network. In addition, some researchers [34], [35] proposed to estimate additional high-quality depth maps to serve as complements to the original depth maps, which is a promising way to improve the quality of depth maps as a whole and alleviate the influence of low-quality depth maps. For saliency inference, the spatial information of the salient regions is more important. However, previous methods weight or improve the entire depth map and did not pay more attention on the salient region. To obtain reliable depth map for SOD, a depth map enhanced algorithm is proposed. We assess the quality of depth information by comparing the performance of SOD with different input modalities. Then, the high-quality samples are used to jointly model depth estimation and salient detection, where the learning of spatial distribution under saliency attention that can provide more reliable depth information for SOD.

In this work, we proposed a new depth-guided fusion network framework (DGFNet) with two-stream encoder and interactive decoder for RGB-D SOD. our main contributions are as follows:

1) We propose a depth map enhanced algorithm. This algorithm uses spatial distribution learning with saliency attention to improve the quality of the depth map.

2) We propose a depth-guided fusion module DGF to fuse RGB and Depth information efficiently, where the spatial attention mechanism aims at reducing the information redundancy, and the weighting mechanisms focuses on guiding the representation of cross-modality features on saliency regions.

3) The results of comprehensive experiments on seven benchmark datasets indicate that our DGFNet is competitive with 23 published RGB-D SOD models.

II. Related Work

Traditional RGB-D SOD with handcrafted features have been widely studied, the performance and generalization of these methods [9], [10], [11] are restricted by the necessity of handcraft feature extraction. Recently, CNN based RGB-D SOD methods have achieved good results. And these methods pay more attention on learning fusion mechanisms of saliency detection in a data-driven manner. The early fusion methods focus on the input-side or the output-side. The input-side fusion [18], [19] directly concatenates the input pair RGB and depth images as a four-channel input information. The output-side fusion [20], [21], [22] predicts saliency results with using each RGB and depth independently, and then fuse two results to generate the final saliency map. These methods cannot effectively fuse cross-modal information due to not considering the macro-level or micro-level CNN features of salient objects.

Feature-Level Fusion Based RGB-D SOD Methods: The feature-level fusion methods simultaneously considered the complementary fusion of cross-modal and cross-level, which becomes a dominant solution in recent years. Chen et al. [23] designed a progressively complementary-aware fusion network to fuse multi-level features from different modalities for locating salient objects. Ji et al. [24] designed a cascaded hierarchical feature fusion strategy to promote efficient information interaction of multi-level contextual features. According to the different feature contributions for prediction from two modalities on different levels, Zhang et al. [27] proposed a top-down multi-level fusion strategy, interweave and gated fusion module were introduced to fuse higher and lower features respectively. Zhang et al. [28] proposed a cross-modality discrepant interaction tetwork (CDINet) for RGB-D SOD. Lower levels of RGB features and higher levels of depth features were used to enhance the details and global features of corresponding modality. Recently, some attention mechanism have been used for feature fusion too. Liu et al. [25] proposed a self-mutual attention module to capture complementary attention knowledge in long-range context propagation to explore the complementary fusion of RGB and Depth. Fan et al. [26] utilized a depth-enhanced module for channel and spatial enhancement of the depth features, and complementary fusion of the enhanced features and the RGB features. Zhou et al. [29] designed a fusion module that learns cross-modal and cross-level features simultaneously, and explores the semantic information. Although the above methods achieved good results, they did not fully utilize the depth data, and fused cross-modal information indiscriminately. Our proposed DGF module uses the spatial information of depth map to enhance the spatial representation of cross-modal features.

Quality of depth map in RGB-D SOD methods: Owing to the quality of the dept map is easily influenced by the sensor, some researchers studied the effects of depth map on SOD and proposed different solutions. Zhao et al. [36] proposed a contrast loss to enhance the contrast of depth map and improve their quality. Liu et al. [37] analyzed the impact of depth quality on the SOD task and proposed a dataset with high-quality depth maps. Chen et al. [30] leveraged the retrieval of a small

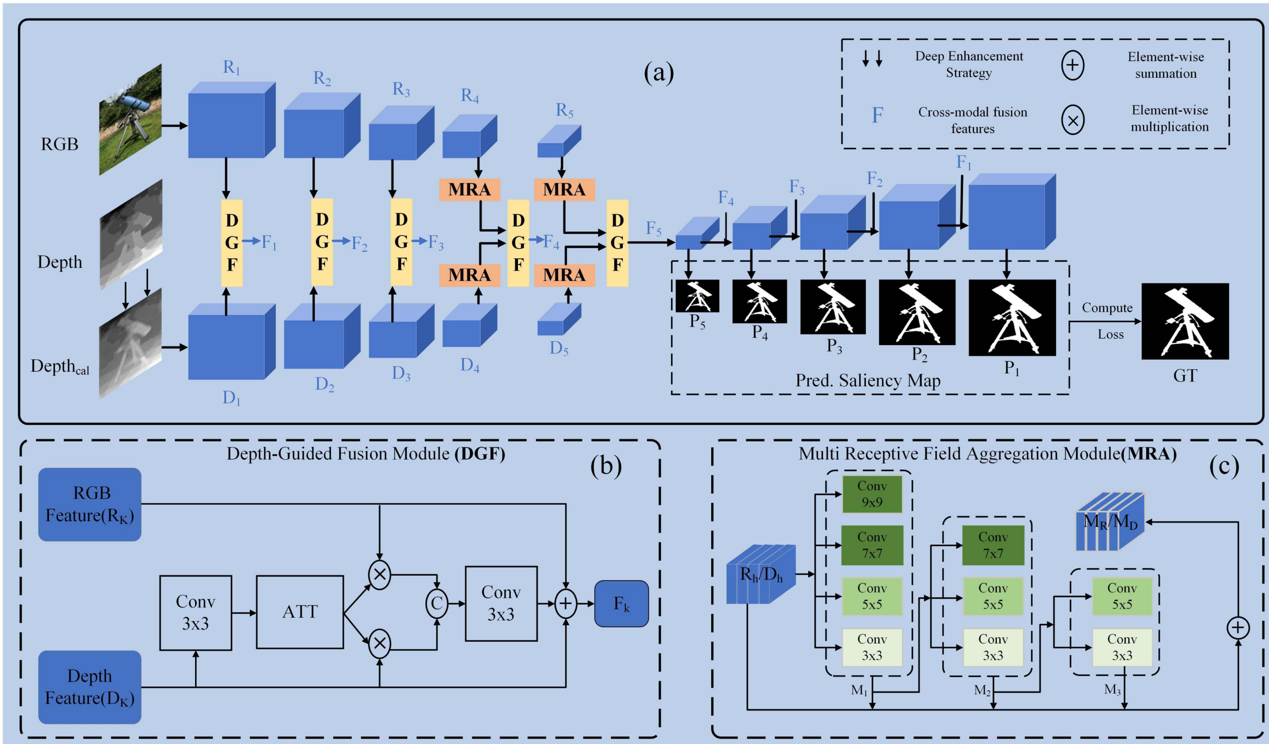

Fig. 2. The overall architecture of our proposed framework. (a) is the process of the Depth-guided fusion network framework (DGFNet). (b) and (c) demonstrate the detailed structure of depth-guided fusion (DGF) and multi-receptive field aggregation (MRA) module respectively.

set of similar images from external datasets to acquire additional enhanced depth information, and employed a selective fusion way to extract hand-crafted saliency clues from the enhanced depth, original depth and RGB image to predict saliency. Chen et al. [32] proposed a weighted cross-modal feature fusion method by introducing perception of the depth potentiality in a learning-based manner. Fan et al. [31] designed a depth depurator unit to automatically discard low-quality depth maps and proposed a human-centered RGB-D data set obtained from real smartphones. Piao et al. [33] remove the depth stream and allow the RGB stream to learn the depth information on knowledge distillation techniques. Piao et al. [35] estimates a supplementary depth map from the input RGB image, and dynamically fuse the features from the original and estimated depth maps to generate saliency-informative depth features. Ji et al. [34] proposed a depth calibration strategy (CAL), which is a method of removing latent noise from depth information by learning the distribution patterns of high-quality depth maps. The calibrated depth map has improved in quality as a whole. However, the spatial information of the salient regions is more important for SOD, because an ideal depth map should help the model to accurately distinguish salient objects from the background. CAL treated each point of the image equally, and did not pay more attention on the salient regions. Therefore, there is a need for a more comprehensive approach which can focus on the spatial structure correction of salient regions. To solve this problem, we improve CAL and propose a depth map enhanced algorithm. The depth map obtained by this algorithm significantly improves the performance of RGB-D SOD.

III. Proposed Methods

As shown in Fig. 2, the proposed network is a symmetrical two-stream encoder-decoder architecture and the input of the network is an RGB image and a depth map obtained by our depth map enhanced algorithm. First, we use the Multi Receptive Field Aggregation Module (MRA) to process the high-level features of both modalities to obtain richer semantic information. Later, RGB features and depth features at different scales are fused by the Depth-Guided Fusion Module to obtain cross-modal fusion features F in different scales. Finally, these features are progressively refined to generate saliency maps.

A. Encoder With Multi-Receptive Field Aggregation Module

Here, a revised VGG-16 by replacing the last pooling layer with an MRA module was used as a feature extractor. All parameters of VGG were initialized by training of ImageNet. RGB and corresponding enhanced depth information are fed individually into the feature extractor. The multi-layer features of two-stream encoders are denoted by R_i and $D_i (i = 1, 2, 3, 4, 5)$ with different sizes. To simplify the network, these features are fed into the convolutional block to reduce the channel numbers of features.

In feature extraction, more and deeper receptive fields can provide more discriminative features for visual tasks. Therefore, we introduce an MRA module to explore better cross-modality features. Considering the amount of computation and loss of details caused by the multi-layer receptive field applied to the low-level features, we use the MRA to aggregate high-level feature of two modalities to fully explore semantic information. As shown in

Fig. 2(c), the module consists of three different layers, and each of them contains different number of convolutions to achieve richer receptive fields and extract more representative features. Taking the RGB stream as an example, the high-level features $R_h (h = 4, 5)$ are processed through three different layers in sequence and obtain the aggregated features in each layer. In this way, richer semantic information can therefore be obtained by combining features with different receptive fields. The process can be expressed as:

$$M_{1,h} = f_{conv}(cat(f_{3\times3}(R_h), f_{5\times5}(R_h), f_{7\times7}(R_h), f_{9\times9}(R_h)))$$
$$M_{2,h} = f_{conv}(cat(f_{3\times3}(R_h), f_{5\times5}(R_h), f_{7\times7}(R_h)))$$
$$M_{3,h} = f_{conv}(cat(f_{3\times3}(R_h), f_{5\times5}(R_h))) \quad (1)$$

Finally, to preserve the original feature information in the aggregated multi-sensory features, $R_h (h = 4, 5)$ are element-wise summed with the aggregated features $M_{i,h}$ of each layer in the form of residual concatenation. So, the aggregated RGB features of the h-th level can be obtained in the following way:

$$M_{R,h} = M_{1,h} + M_{2,h} + M_{3,h} + R_h \quad (2)$$

Before cross-modal fusion, we use the MRA module to explore semantic information from high-level features of RGB and depth respectively. The fused aggregation feature will feed to the corresponding decoder layer.

B. Depth Map Enhanced Algorithm

Motivated by [34], we evaluate the quality of the depth maps with the contribution of the depth map to the SOD. In order to improve the quality of depth, we calibrate the original one learning from some high-quality depth maps.

In this article, we consider the result of RGB-D salient objects detection can reflect the quality of the depth map. More reliable depth information can result in more accurate detection. We train two baseline models with the same architecture for $\{rgb_i\}_{i=1}^N$ and $\{depth_{raw,i}\}_{i=1}^N$ individually as input under the supervision of saliency ground-truths $\{GT_i\}_{i=1}^N$, denoted as SN_{rgb} and SN_{depth}, respectively. The baseline model is an encoder-decoder network where the encoder is the ResNet34 and the decoder is the decoder part of U-Net. For each input of training sets $\{rgb_i, depth_{raw,i}\}_{i=1}^N$, the two baseline models can be used to predict the salient results. Then, we get the the intersection over union scores IOU^r and IOU^d between predicted results (P_i^{rgb}, P_i^{depth}) and ground-truths GT_i. Ranking the IOU^d and comparing it with IOU^r can help select the representative sample. Specifically, the samples ranked top 20% of training sets denoted the S_{pos} (the quality of depth map is good for SOD) and the bottom 20% denoted the S_{neg} (the quality of depth map is bad for SOD). In addition, when $IOU^d > IOU^r$, these samples denoted the S_{pos}, which demonstrates the depth map makes more contribution than the RGB image for SOD.

In order to evaluate the quality of the depth map, we also use the $\{depth_{raw}, 1\} \in S_{pos}$ and $\{depth_{raw}, 0\} \in S_{neg}$ sets to train a ResNet-18 based discriminator network for two classification tasks. For each input raw depth information, we can use the trained network to give the reliability score g which indicates the probabilities of the depth map being positive. The predicted score can reflect the reliability of the original depth map.

To estimate depth information, we design a dual-branch framework consisting of a SOD branch and a depth estimation branch. The network shares the same feature extractor to encode the input RGB image $\{rgb_{raw}\} \in S_{pos}$ and supervised with $\{depth_{raw}, GT\}$. The SOD branch is guided with salient ground truth of the positive samples, which can be used to identify salient regions and get the salient spatial attention. The depth estimation branch was trained with positive depth samples, which can be used to get more accurate depth information with the help of the saliency attention. In this way, the trained dual-branch model can be used to predict saliency masks and estimate more reliable depth maps $depth_{est}$ with only RGB information.

Finally, the enhanced high-quality depth map $depth_{cal}$ is obtained by weighted fusion of $depth_{est}$ and raw depth map with g. The depth map enhanced algorithm is further explained in Algorithm 1. It should be noted that the proposed model does not require additional any supervised information except for the normal SOD training datasets. To avoid the influence of low-quality depth maps on salient detection, we employ this algorithm to improve the quality of the depth maps in all the training and test sets.

C. Depth-Guided Fusion Module

Since RGB image and depth map are gathered from multiple sources in different aspects, which provide complementary information for scene analysis. For the enhanced depth map, its spatial information is helpful for localize and segment the salient object. To effectively utilize the spatial information of depth maps and integrate complementary information across modalities, a depth-guided fusion DGF module is proposed. Firstly, we generate a set of attention maps for each level with the extracted feature of the calibrated depth. The attention maps represent the spatial structure information and used as distribution maps to weight RGB and Depth features. And then the enhanced feature of RGB and Depth modalities are fused to obtain the cross-modality fusion feature.

Specifically, as shown in Fig. 2(b), we obtained the cross-modal complementary features by the depth-guided manner. For each level, the features $D_k (k = 1, 2, 3, 4, 5)$ from the depth stream is firstly processed by applying two convolutional layers $f_{conv3\times3}$ to generate the smoothed feature. Then, the smoothed feature is analyzed by one convolutional block $conv_{3\times3}$ and the sigmoid activation function S to obtain the attention map Att_k. The process can be expressed as follows:

$$Att_k = S(conv_{3\times3}(f_{conv3\times3}(f_{conv3\times3}(D_k)))) \quad (3)$$

Then, the RGB features $R_k (k = 1, 2, 3, 4, 5)$ and D_k depth features are element-wise multiplied with the corresponding Att_k to focus on the desired part of features. In this way, the spatial structure of cross-modality features can be enhanced and highlighting the representation of salience cues. At the same time, the depth attention maps give lower weight to irrelevant features. It suppresses the expression of irrelevant features and

Algorithm 1: Depth Map Enhanced

```
Input: {rgb_i, depth_raw,i, GT_i}_{i=1}^N
Output: depth_cal,j
// Selecting the representative sample
Training the network SN_rgb and SN_depth with
   {rgb_i, GT_i}_{i=1}^N and {depth_raw,i, GT_i}_{i=1}^N, respectively.
Get the saliency predictions P_i^rgb = SN_rgb(rgb_i) and
   P_i^depth = SN_depth(depth_raw,i)
Calculating the IOU of {P_i^rgb, P_i^depth, GT_i} and get
   IOU^r and IOU^d with sorting
Set S_pos = ∅ and set S_neg = ∅
for i = 1 to N do
   if IOU_i^d ∈ the top 20% of IOU^d or IOU_i^d > IOU_i^r
   then
      S_pos = S_pos ∪ (rgb_i, depth_raw,i, GT_i)
   end if
   if IOU_i^d ∈ the bottom 20% of IOU^d then
      S_neg = S_neg ∪ (rgb_i, depth_raw,i, GT_i)
   end if
end for
// Evaluating the reliability of the original depth map
Training a discriminative network with
   {depth_raw, 1} ∈ S_pos and {depth_raw, 0} ∈ S_neg
For the depth_raw,j in the training sets and test sets, get
   the reliability score g_j of with the trained discriminative
   model
// Obtaining the new depth information
Training the dual-branch prediction network for
   {rgb_raw} ∈ S_pos as input under the supervision of
   {depth_raw, GT} ∈ S_pos
Estimating depth map depth_est,j by the trained model
// Calibrating the original depth map
   For any raw depth map depth_raw,j :
depth_cal,j = depth_est,j * (1 − g_j) + depth_raw,j * g_j
Return depth_cal,j
```

reduces the gap between RGB and depth at the feature level. These enhanced features are concatenated along the channel and then feed into the convolutional layer to obtain the fused features F_k. Finally, to preserve the original feature representation, we update the F_k with the original RGB and depth features through channel-concatenation. The process of fusion can be expressed as:

$$F_k = f_{conv3\times3}(cat(f_{conv3\times3}(cat(R_k * Att_k, D_k * Att_k)), R_k + D_k)). \tag{4}$$

Since complementary information exists not only in the higher-level features but also in the lower-level features. For the features D_k and R_k, we use DGF modules to obtain cross-modal complementary features F_k at each level. In the decoder, followed by a transition convolution block $conv_{3\times3}$ and a sigmoid layer S, the features F_k are channel-spliced with the last level features F_{k+1} with up-sampling to generate the saliency maps P_k. In this way, the cross-model fusion features $F_k (k = 5, 4, 3, 2, 1)$ are progressively refined to generate five saliency map with different scales $(P_5, P_4, P_3, P_2, P_1)$. The process of saliency inference can be denoted as:

$$P_k = S(conv_{3\times3}(cat(F_k, up(F_{k+1})))) \tag{5}$$

D. Training Loss

In the training of our DGFNet, the total loss function is:

$$L_{sod} = \sum_{k=1}^{5} (L_{bce,k} + L_{dc,k}) \tag{6}$$

For each level output, both $L_{bce,k}$ and $L_{dc,k}$ are used for measuring the loss between the prediction map and ground truth with different scales. L_{bce} and L_{dc} represent the binary cross-entropy loss [38] and the dice coefficient loss [39], which are widely used in segmentation tasks. We upsample the output P_k to the original input size, and calculate the loss with the ground truth GT.

IV. Experiments

A. Implementation Details

We used PyTorch to implement our DGFNet. The model was trained with 200 epochs and updated the network parameters using Adam optimizer. We set batch size as 10, learning rate as 1e-4 with reduced to 1/10 times every 60 iterations. Data augmentation was performed to enlarge the dataset via flipping, rotation and random clipping. To feed RGB-D image pairs into the network, for both the training and testing processes, we resized the images to 224*224 pixels.

B. Dataset

We experimented on 7 public RGBD salient object detection datasets, including DUTLF-D [40], LFSD [41], NJU2K [42], NLPR [43], STERE [44], DES [45] and SIP [37]. NJU2K contains 2003 RGB-D image pairs, which have been obtained from the Internet, 3D movies, stereo camera, etc. STERE includes 1000 pairs of RGB-D image pairs from the Web. DES is also known as RGBD135 which contains a total of 135 image pairs obtained by Kinect. SIP is a newly proposed dataset that includes 929 annotated high-resolution images. DUTLF-D [24] is a recently popular dataset for RGB-D SOD, and it consists of 1,200 images from 800 indoor and 400 outdoor scenes, which are challenging and complex. LFSD contains 100 images with depth information captured via a Lytro light field camera. Following the dataset splitting strategy of [31], [46], [47], we use 2185 samples consisting 1485 samples of NJU2K and 700 samples of NLPR as the training sets, and the remaining samples served as the test sets. In addition, we used 800 RGB-D images of DUTLF-D for training and the 400 remaining RGB-D images for testing.

C. Evaluation Metric

To evaluate the predicted saliency maps, we adopt four widely used evaluation metrics: maximum F-measure [48] (maxF), maximum E-measure [49] (maxE), structural similarity [50]

TABLE I
QUANTITATIVE COMPARISON WITH DIFFERENT ABLATION SETTINGS

Index	RGB	$Depth_{raw}$	$Depth_{cal}$	DGF	CAT	SUM	MRA	NJU2K S-m	NJU2K Emax	NJU2K Fmax	NJU2K MAE	STERE S-m	STERE Emax	STERE Fmax	STERE MAE
a	✓							0.897	0.930	0.892	0.043	0.878	0.886	0.872	0.043
b		✓						0.862	0.905	0.851	0.062	0.703	0.797	0.655	0.121
c			✓					0.889	0.919	0.889	0.047	0.865	0.841	0.869	0.054
d	✓	✓		✓				0.913	0.943	0.913	0.035	0.896	0.939	0.892	0.040
e	✓	✓		✓			✓	0.915	0.948	0.916	0.034	0.897	0.938	0.893	0.041
f	✓		✓		✓			0.910	0.943	0.912	0.037	0.901	0.941	0.897	0.039
g	✓		✓			✓		0.909	0.941	0.908	0.038	0.897	0.940	0.892	0.040
h	✓		✓	✓				0.918	0.945	0.919	0.033	0.908	0.946	0.906	0.036
j(Ours)	✓		✓	✓			✓	**0.921**	**0.953**	**0.925**	**0.032**	**0.911**	**0.948**	**0.911**	**0.035**

(S-measure) and mean absolute error [51] (MAE). Specifically, the F-value provides a more balanced evaluation between mean precision P and mean recall R over saliency maps. It can be denoted as:

$$F_\beta = \frac{(1+\beta^2)\times P \times R}{\beta^2 \times P + R} \tag{7}$$

where β^2 is set to 0.3 as usual.

The E-value is an enhanced alignment measure, which combines local-pixel values and image-level mean in one term to capture visual matching information.

$$E = \frac{1}{W \times H}\sum_{i=1}^{W}\sum_{j=1}^{H}\phi_s(i,j) \tag{8}$$

where ϕ_s is the enhanced alignment matrix, which reflects the correlation between salient map S and ground truth G by subtracting their global means, respectively. The S-value jointly evaluated the structural similarity between the object perception S_o and regional perception S_r as follows:

$$S = \alpha * S_o + (1-\alpha)S_r \tag{9}$$

where α is the balance parameter and is set to 0.5. The MAE calculates mean pixel-wise absolute difference between the predicted values and the ground truth salient map.

$$MAE = \frac{1}{W \times H}\sum_{i=1}^{W}\sum_{j=1}^{H}|S(i,j) - G(i,j)| \tag{10}$$

where S is a normalized saliency map, W and H denote the width and height of S. For comparison, the lower the value of MAE is, the better the performance will be. For other metrics, the higher score is better.

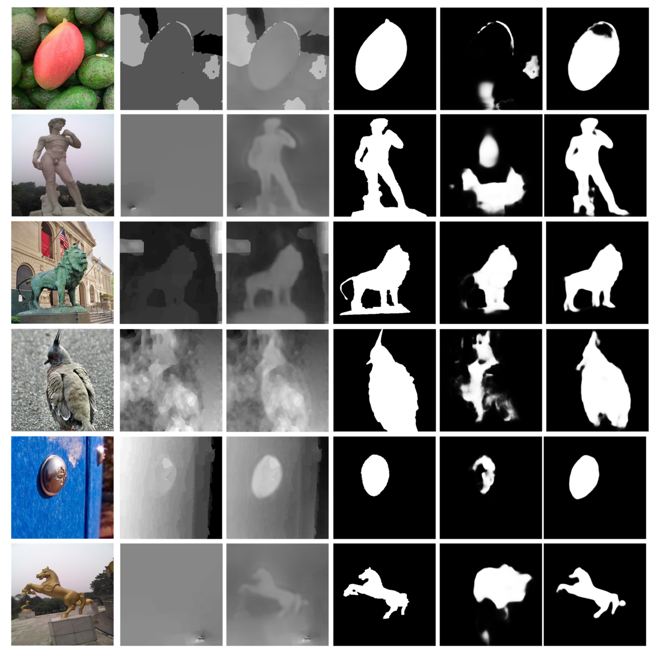

Fig. 3. Visual comparisons of different depth maps. The first column is the RGB image, the second and third columns individually are the original depth map $depth_{raw}$ and the $depth_{cal}$ obtained by our proposed algorithm. The fourth column is the GT, the fifth and sixth columns are the saliency results from $depth_{raw}$ and $depth_{cal}$, respectively.

D. Ablation Analysis

To validate the effectiveness of the proposed module, we designed multiple ablation experimental schemes on two datasets, NJU2K and STERE.

Effectiveness of the depth map enhanced algorithm: To improve the quality of the depth map, we use the depth map enhanced algorithm with some saliency cues from selected positive samples. To demonstrate its effect, we compare the performance of single-stream networks with only one modality as input, including RGB, corresponding raw depth, enhanced depth information. We validate the results in all benchmarks following a common training strategy. The results show in the first three rows of Table I. Compared to the results with the raw depth maps, we can see that using enhanced depth map achieves much better performance, which increased S-measure by 3.1%, 23%, maximal E-value by 1.5%, 5.5%, upgrade 4.4%, 32% in terms of the F-max and decreased the MAE by 24%, 55% on datasets NJU2K and STERE respectively. The results show that the enhanced depth can get the comparable results with normal RGB. Also, we give some visual comparison in Fig. 3. Compared to the second and third columns, we can conclude that the enhanced depth images $depth_{cal}$ keep more accordance with the GT. Compared to the saliency results from these depth map in the fifth and sixth columns, we can find that the enhanced depth images $depth_{cal}$ is more contributive in salient object detection. The

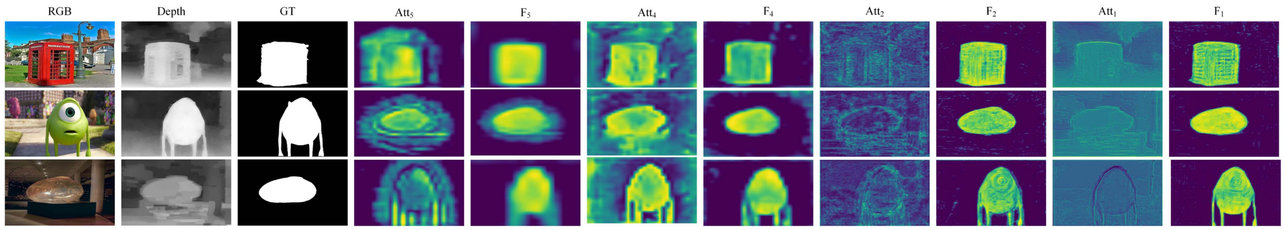

Fig. 4. Visual results of cross-modal fusion in lower and higher level.

TABLE II
GENERALIZATION CAPABILITY OF THE PROPOSED METHOD

Model	NJU2K			STERE		
	S-m	Fmax	MAE	S-m	Fmax	MAE
DMRA [24]	0.886	0.872	0.051	0.886	0.875	**0.047**
DMRA+$depth_{cal}$	**0.903**	**0.879**	**0.046**	**0.898**	**0.876**	**0.047**
BBSNet [26]	0.923	0.916	0.037	0.911	0.909	0.039
BBSNet+$depth_{cal}$	**0.926**	**0.924**	**0.034**	**0.916**	**0.911**	**0.036**
DGFNet+$depthst$	0.918	0.916	0.033	0.907	0.895	0.037
DGFNet+$depth_{cal}$	**0.921**	**0.925**	**0.032**	**0.911**	**0.911**	**0.035**
DGFNet*+CDC	0.914	0.918	0.035	0.907	0.904	**0.036**
DGFNet*+DRB	0.910	0.913	0.036	0.902	0.902	0.038
DGFNet*+DGF	**0.918**	**0.919**	**0.033**	**0.908**	**0.906**	**0.036**

learning saliency cues enable the module pay more attention on saliency regions, which can efficiently distinguish foreground from some complex background.

The fifth and sixth rows of Table II show the RGB-D results with two different depth enhancement schemes, the $depthst$ was obtained by the similar calibration method [34]. The comparison results show that the depth map obtained by our method works better. Furthermore, to evaluate the generalization capability of our proposed algorithm, we revise some state-of-art RGB-D SOD methods, including DMRA and BBSNet. The input depth information of the revised methods was replaced with our $depth_{cal}$. The evaluation results are presented in rows 1 to 4 of Table II. From the third and fourth rows, we can see the performance improvement on two datasets in all three metrics. The results suggest that the spatial attention is more helpful for calibrating depth information.

Effectiveness of the DGF module: In order to evaluate the effectiveness of the depth-guided fusion module, we compare the performance of normal RGB-D feature fusion and our DGF method. The evaluation results are presented in rows "f", "g" and "h" of Table I. Compared to the normal fusion scheme with addition operations, we observed that the proposed DGF obtained superior detection results, which improved 0.9%, 1.2% in terms of the S-m, 1.2%, 1.5% in terms of the F-max and decreased the MAE by 13%, 10% on datasets NJU2K and STERE. Compared to the fusion scheme with convolution operations, we can see that our method also obtained better results, which improved 0.7%, 1.0% in terms of the F-max and decreased the MAE by 10%, 7.6% on datasets NJU2K and STERE. Additionally, we use some other fusion method to replace the DGF in our framework and the comparison results are shown in the last three rows of Table II. DGFNet* is our baseline with the $depth_{cal}$. CDC and DRB are the fusion methods from [52] and [24], respectively. The comparison results show that the proposed DGF helps our network to achieve superior performance. The DGF module has been used in five cross-modality fusion stages, we visualize some fusion results in Fig. 4. Columns 4 to 7 show higher-level fusion features and attention maps, which provide the coarse localization of salient objects. As the fusion goes through, the regularized features become depth-aware, and boundaries also become progressively clearer. As shown in the 8 to 11 columns, the lower-level fusion features and the attention maps contain more details, and help refine the prediction results gradually. These above indicate that the proposed DGF can better explore complementary information across modalities.

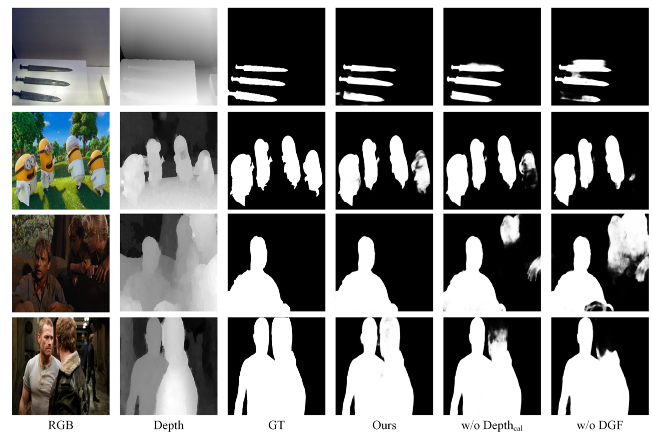

Fig. 5. Visual examples of saliency maps of each module.

Contribution of each module: To visually present the characteristics of each module, some examples are visualized in Fig. 5. We could observe that the DGF helps to recognize and locate the salient object while $Depth_{cal}$ provides better detail.

Besides the two key observations above, we also design the ablation experiments with the MRA. The evaluation results are presented at the rows "d", "e","h" and "j" of Table I. Comparing the d and e rows, we can see that the method with MRA obtained improved performance on datasets NJU2K and STERE. The comparison between h and last rows demonstrates that our method also achieves better performance after using the MRA. These improvements validate the effectiveness of MRA for salient object detection. In conclusion, the DGF and MRA modules are effective for the model. From performance improvement and visualization, we can see that the DGF module plays a major role in the improvement of the model performance. Furthermore, comparing the results of d,e,h and last rows of Table I, we find that the proposed modules are also effective for the original depth map.

TABLE III
QUALITATIVE COMPARISON OF THE PROPOSED APPROACH WITH 23 STATE-OF-THE-ART RGB-D SOD METHODS

year&journal	Model	DUTLF-D			LFSD			NJU2K			NLPR			DES			STERE			SIP		
		S-m	F-m	MAE	S-m	F-m	MAE	S-m	F-m	MAE	S-m	F-m	MAE	S-m	F-m	MAE	S-m	F-m	MAE	S-m	F-m	MAE
2019ICME	PDNet [53]	-	-	-	-	0.883	0.076	0.835	0.740	0.064	0.886	0.795	0.041	0.868	0.800	0.050	0.874	0.833	0.064	-	-	-
2019CVPR	TANet [54]	0.808	0.779	0.093	0.801	0.794	0.111	0.878	0.844	0.061	0.886	0.795	0.041	0.858	0.782	0.045	0.877	0.849	0.060	-	-	-
2019CVPR	CPFP [36]	0.749	0.736	0.099	0.828	0.813	0.088	0.878	0.850	0.053	0.888	0.822	0.036	0.874	0.819	0.037	0.871	0.827	0.054	0.852	0.863	0.060
2020ECCV	BBSNet [26]	0.682	0.663	0.120	0.843	0.831	0.081	0.912	0.893	0.040	0.920	0.867	0.027	0.906	0.855	0.029	0.892	0.860	0.048	0.871	0.851	0.057
2020TNNLS	D3Net [31]	0.736	0.756	0.097	0.831	0.836	0.099	0.895	0.889	0.051	0.906	0.885	0.034	0.902	0.885	0.031	0.891	0.881	0.054	0.862	0.862	0.063
2021CVPR	CALNet [34]	**0.836**	**0.821**	**0.070**	0.841	0.841	**0.074**	0.911	**0.902**	**0.035**	0.923	0.891	**0.021**	0.905	0.893	0.023	0.901	**0.885**	**0.038**	0.874	**0.875**	**0.051**
2021TCSVT	SNet(VGG$_{16}$) [56]	-	-	-	-	-	-	-	-	-	0.911	**0.894**	0.039	0.925	0.878	0.026	0.896	0.872	0.044	**0.892**	**0.879**	**0.046**
2021TCSVT	EBFNet [66]	-	-	-	-	-	-	0.904	0.891	0.039	0.916	**0.893**	0.025	-	-	-	0.900	0.868	0.043	**0.886**	0.867	0.049
2021TIP	BIANet [13]	-	-	-	-	-	-	0.915	0.892	0.039	0.925	0.876	0.025	**0.931**	**0.902**	0.021	0.904	0.871	0.043	0.882	0.872	0.053
2021TIP	BSNet [60]	-	-	-	**0.864**	**0.857**	**0.072**	**0.920**	**0.899**	**0.035**	**0.930**	0.877	**0.023**	0.933	0.891	0.021	**0.908**	**0.882**	0.041	0.878	0.869	0.055
2021TIP	CDNet [35]	**0.885**	**0.862**	**0.049**	**0.858**	**0.855**	**0.073**	0.912	0.891	0.038	**0.930**	0.873	0.025	**0.937**	0.885	**0.019**	0.903	0.873	**0.039**	0.861	0.847	0.060
2021TIP	HaiNet [58]	-	-	-	-	-	-	**0.913**	0.883	0.038	0.924	0.887	0.024	**0.935**	**0.910**	**0.018**	**0.907**	0.866	0.040	0.880	0.842	0.053
Ours$_{NJU2k+NLPR}$		**0.880**	**0.862**	**0.054**	**0.848**	**0.846**	0.075	**0.921**	**0.914**	**0.032**	**0.928**	**0.902**	**0.021**	0.926	**0.912**	**0.019**	**0.911**	**0.896**	**0.035**	**0.883**	**0.879**	**0.048**
2020CVPR	SFF [46]	0.915	0.915	0.033	0.859	0.867	0.066	0.899	0.886	0.043	0.914	0.875	0.026	0.905	0.876	0.025	0.893	0.880	0.044	-	-	-
2021CVPR	CALNet [34]	0.924	0.923	0.029	0.855	0.857	0.070	0.903	0.897	0.038	0.921	0.892	0.023	0.917	0.894	0.021	0.905	0.888	**0.036**	0.872	0.877	0.052
2021CVPR	RD3D [55]	0.931	0.920	0.029	0.857	0.854	0.738	0.915	**0.898**	0.037	0.921	0.886	**0.022**	**0.939**	0.904	0.019	**0.911**	0.882	0.037	**0.884**	0.871	**0.049**
2021ACM MM	DFMNet [59]	0.912	0.892	0.037	0.865	0.858	0.072	0.907	0.882	0.043	0.922	0.865	0.026	0.933	0.884	0.021	0.898	0.856	0.043	0.882	0.870	0.051
2021ACM MM	TriTransNet [57]	**0.932**	**0.938**	**0.025**	-	-	-	**0.920**	**0.919**	**0.031**	**0.928**	**0.909**	**0.020**	**0.943**	**0.936**	**0.014**	0.886	**0.892**	0.043	0.877	**0.881**	0.054
2021TIP	CDNet [35]	0.927	0.920	0.029	**0.875**	**0.873**	**0.062**	0.916	0.897	0.037	0.930	0.880	0.024	**0.937**	0.892	0.019	**0.910**	0.882	**0.036**	0.874	0.865	0.053
2021TIP	HaiNet [58]	0.910	0.883	0.038	-	-	-	0.909	0.879	0.038	0.921	0.881	0.025	0.929	0.898	0.019	0.909	0.871	0.038	**0.886**	0.854	**0.048**
2021TMM	CCAFNet [29]	0.905	0.907	0.036	0.827	0.832	0.087	0.910	0.898	0.037	0.922	0.882	0.026	0.930	**0.916**	**0.018**	0.891	0.870	0.044	0.877	0.866	0.054
2022TCSVT	RGDNet [62]	0.927	-	0.031	0.865	-	0.068	**0.917**	-	**0.036**	0.921	-	0.026	0.934	-	0.019	0.900	-	0.043	0.806	0.819	0.085
2022TMM	C2DFNet [65]	**0.933**	**0.934**	**0.025**	0.863	0.863	**0.065**	0.907	0.899	0.038	0.921	**0.899**	**0.021**	0.923	0.907	0.019	0.901	**0.892**	0.038	0.870	**0.881**	0.052
2022TIP	DMRA [24]	0.888	0.883	0.048	0.847	0.849	0.075	0.886	0.872	0.051	0.899	0.855	0.031	0.901	0.857	0.029	-	-	-	0.806	0.819	0.085
2022TIP	CIRNet [64]	0.928	0.903	0.033	**0.877**	0.861	0.068	0.915	0.895	0.039	**0.929**	0.869	0.025	-	-	-	-	-	-	0.879	0.863	0.055
2022TIP	DCMF [63]	0.928	0.912	0.032	0.868	**0.870**	0.069	0.908	0.867	0.045	0.921	0.849	0.029	0.935	0.882	0.022	0.909	0.864	0.041	0.806	0.819	0.085
Ours$_{NJU2k+NLPR+DUT}$		**0.932**	**0.925**	**0.027**	**0.869**	**0.865**	**0.065**	**0.920**	**0.912**	**0.033**	**0.923**	**0.893**	0.025	0.931	**0.916**	**0.018**	**0.913**	**0.894**	**0.035**	**0.892**	**0.888**	**0.044**

E. Comparison to State-of-The-Art Models

We compared our method with 23 SOTA methods: PDNet [53], CPFP [36], TANet [54], D3Net [31], SFF [46], BBSNet [26], RD3D [55], CalNet [34], SNet [56], TriTransNet [57], CDNet [35], HaiNet [58], BIANet [13], DFMNet [59], BSNet [60], CCAFNet [29], CMINet [61], DMRA [24], RGDNet [62], DCMF [63], CIRNet [64], LDNet [63], C2DFNet [65]. These methods learned hierarchical features under a deep learning framework. For a fair comparison, we used all saliency maps provided by the authors or obtained by the given codes. We evaluated them with the same evaluation measures in quantitative comparison and qualitative comparison.

Quantitative comparison: Table III shows the results of related methods on some widely-used RGBD salient object detection datasets (DUTLF-D, LFSD, NJU2K, NLPR, DES, STERE, and SIP). Following the dataset splitting strategy [46] and [24], two different training settings(NJU2K+NLPR and NJU2K+NLPR+DUTLF-D) are adopted, the results of which are independently listed in the first and second block of Table III. From the Table, we can see that the proposed method outperforms the other 23 methods in the overall state. On the seven datasets, our method achieves the first, second or third ranking. The comparison results demonstrate the effectiveness of the present method. Especially, the DGFNet shows superior performance compared to other methods on the STERE dataset, where the images are comparably complicated. This further indicates that our model can take full advantage of the depth map to deal with the complex scenes more effectively. In Table IV, we compare the number of parameters and Flops with the state-of-the-art RGB-D salient object detection methods. our method has small parameters and acceptable flops. Especially TriTransNet [57], our method does not have the best performance, but lower complexity.

Qualitative comparison: To further illustrate the advantages of the present method, we visually compare our method with other five RGB-D SOD methods in Fig. 6. From the results, we can see that the saliency maps obtained by our method are much more consistent with the ground truth. Specifically, for some poor depth information, as shown in the 1st and 8th rows, we can find that some compared methods TANet [54], SFF [46], D3Net [31], CPFP [36] and PDNet [53] fail to detect edges or boundaries of salient regions. The detection result of our method shown in the fourth column is greatly improved compared with other methods. The 5th row gives performance comparison of different models for dealing with images containing multiple targets. The other models have difficulty to distinguish the contours of multiple saliency areas in a same scene, while our method performs very well. As shown in the 6th row, our method also can get the complete salient object and good details for analyzing image with a small object. While most of the existing methods failed to this kind of challenging scene. In addition, we also give the visual comparison for dealing with low-contrast images (shown in the 3 rd and 7th rows). The proposed method performs well because the detection algorithm can accurately detect and locate entire conspicuous objects with sharp details. In addition, we show some examples in extreme environments. The last row shows the case with ambiguous annotations. Two objects in the image have similar texture and depth values. As DGFNet exploits the depth information to guide the fusion, the saliency maps of DGFNet focus on the object regions more globally. In the second last row, the salient object has complex non-geometric results and the salient results lack fine details. This result show that our approach needs to be enhanced in the details of the complex

TABLE IV
COMPLEXITY ANALYSIS OF THE PROPOSED METHOD WITH OTHER RGB-D SOD METHODS

year&journal	Model	FLOPs(G)	Param.(MB)
2020ECCV	BBSNet [26]	28.682	49.769
2021TIP	HAINet [58]	73.549	59.822
2021CVPR	CalNet [34]	55.476	107.291
2021TIP	CDNet [35]	68.974	53.550
2021TCSVT	SNet [56]	124.723	198.783
2021ACM MM	TriTransNet [57]	293.986	138.761
2022TIP	CIRNet [64]	156.341	82.079
2022TIP	C2DFNet [65]	**22.046**	47.515
DGNet(Ours)		74.885	**42.140**

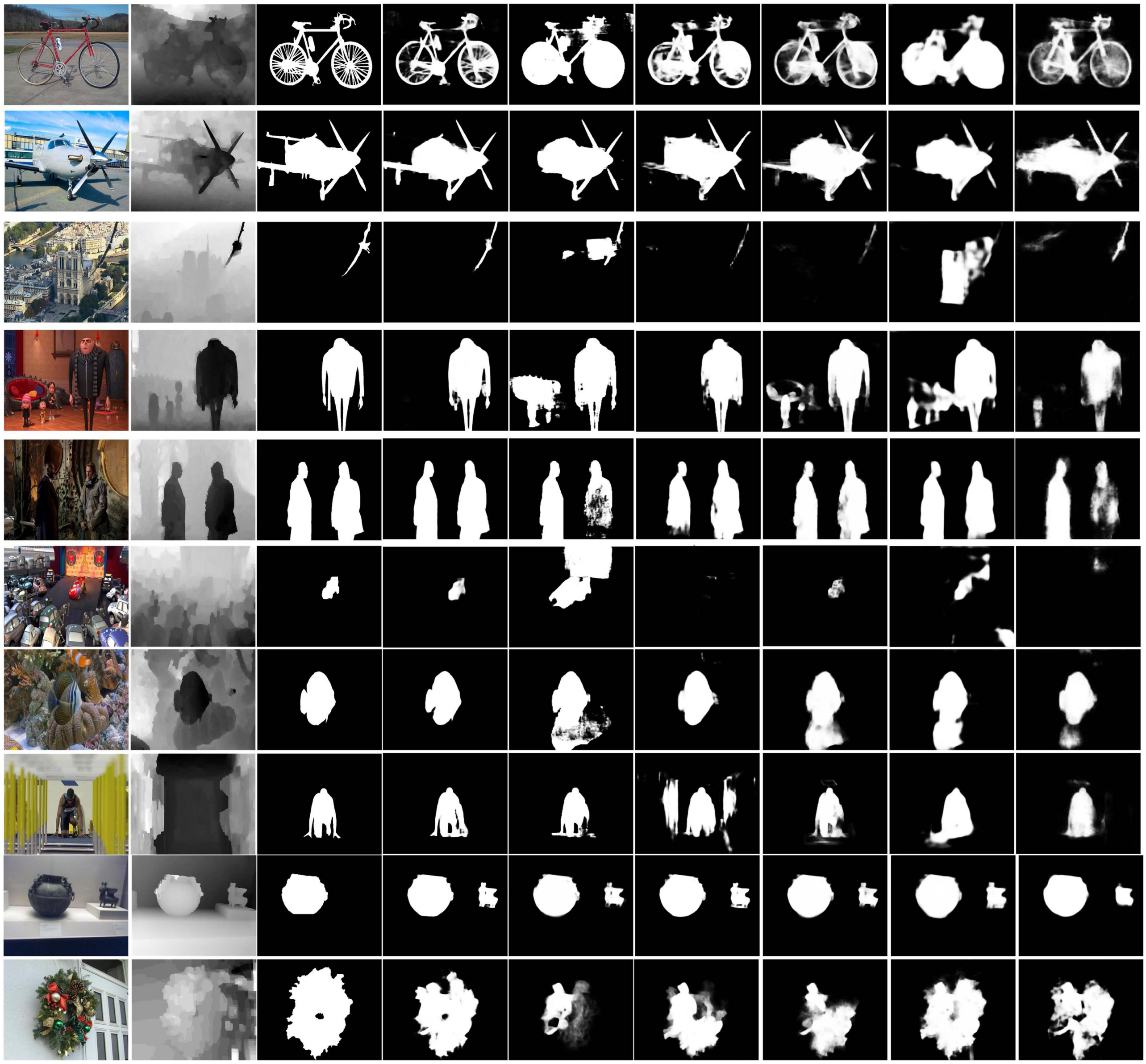

Fig. 6. Visual comparison of the proposed approach with some state-of-the-art RGB-D SOD methods.

structure. Compared to other methods, The detection result of our method is more reliable in these extreme environments.

These qualitative results further indicate the power of the proposed DGFNet to detect salient regions. The proposed method provides new insights into a variety of challenging issues for SOD such as unreliable depth information, complex background and different numbers of salient objects. Also, we found the impact of extreme environments on salient object detection, which guides the direction of our future work.

V. Conclusion and Discussion

In this article, we propose a new network DGFNet for RGB-D SOD. To fully explore the semantic information in high-level features, we proposed an MRA module that utilizes multiple receptive field aggregation to provide reliable global guidelines for cross-modal fusion. To effectively aggregate the cross-model information, we proposed a DGF module to highlight the representation of salience cues and effectively fuse the cross-model features. In addition, to obtain high-quality depth maps, we proposed a depth map enhanced algorithm. The experimental results on seven datasets show that the proposed DGFNet outperforms the other 23 RGB-D SOD methods. In the next work, we will improve the SOD performance in two ways, including enhancing the detail and improving the ability to discriminate salient object.

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